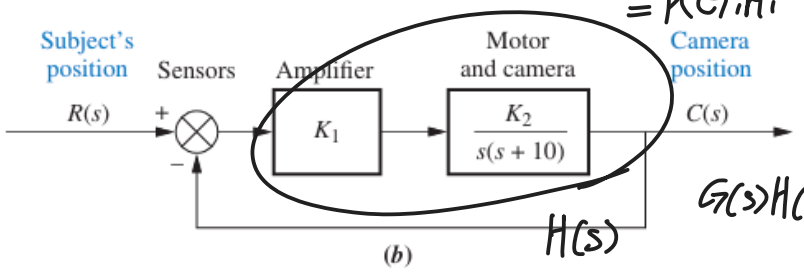
제어공학2
Automatic Control

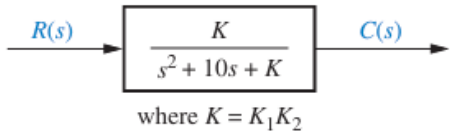
Frequency-Domain Analysis

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근궤적 Loot Locus





K	Pole 1	Pole 2
0	−10	0
5	−9.47	−0.53
10	−8.87	−1.13
15	−8.16	−1.84
20	−7.24	−2.76
25	−5	−5
30	−5 + j2.24	−5 − j2.24
35	−5 + j3.16	−5 − j3.16
40	−5 + j3.87	−5 − j3.87
45	−5 + j4.47	−5 − j4.47
50	−5 + j5	−5 − j5

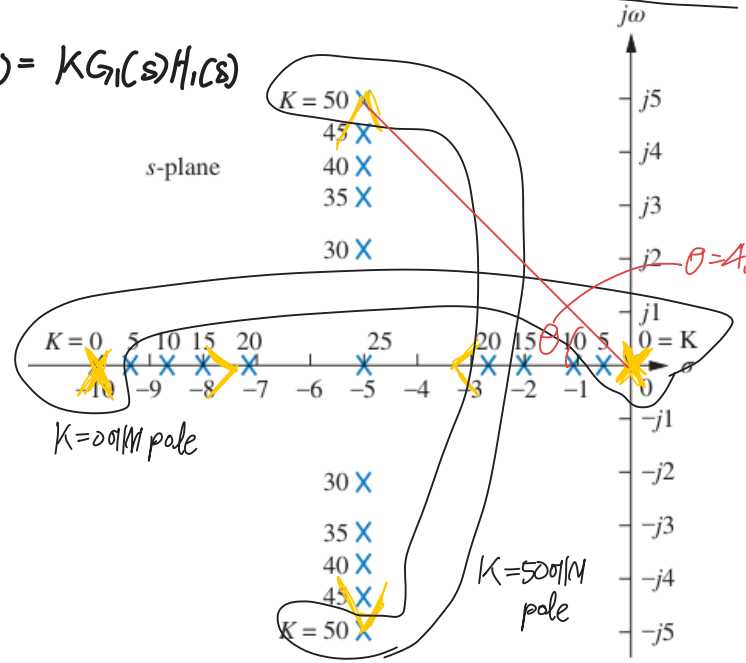
$G(s) = \frac{K_1 K_2}{s(s+10)} = \frac{K}{s(s+10)}$

$\Rightarrow \frac{G}{1+GH} \Rightarrow \frac{\frac{K}{s(s+10)}}{1+\frac{K}{s(s+10)}} \Rightarrow \frac{K}{s^2+10s+K}$

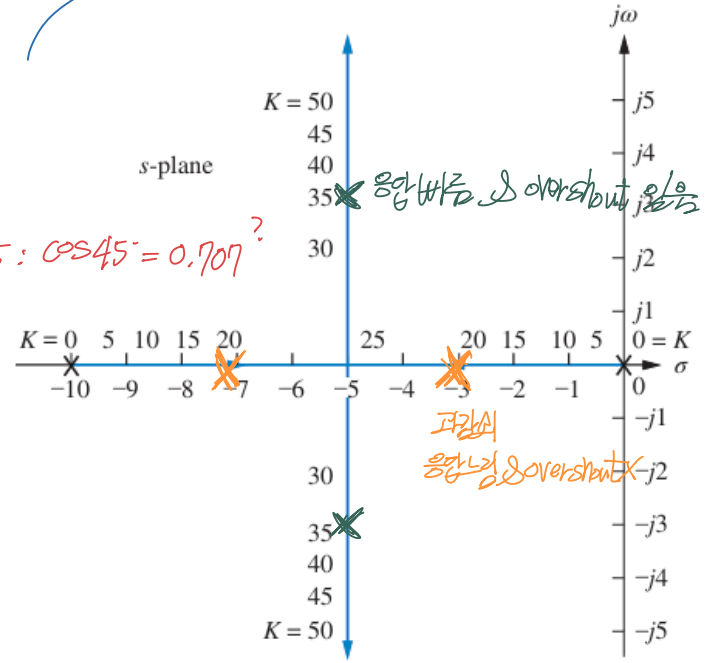
$= KG_1H_1$

$H(s)$

$G(s)H(s) = KG_1(s)H_1(s)$



목적: K를 조절하여 응답속도조절



$\theta = 45 : \cos 45^\circ = 0.707$?

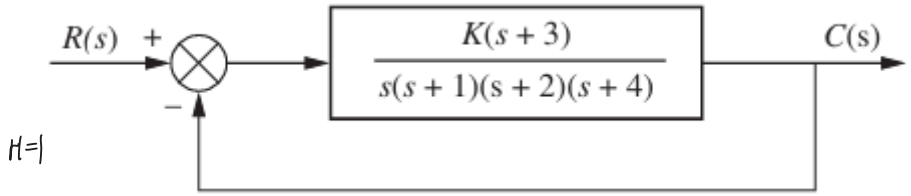
점근선각 : $\theta = \frac{(2i+1)}{|n-m|} \times 180^\circ = 90^\circ, 270^\circ$

$\frac{1}{s(s+10)} = G_1(s)H_1(s)$ $\begin{pmatrix} m=0 \\ n=2 \end{pmatrix}$

$\frac{-10}{n-m} = -5$

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근궤적 Loot Locus

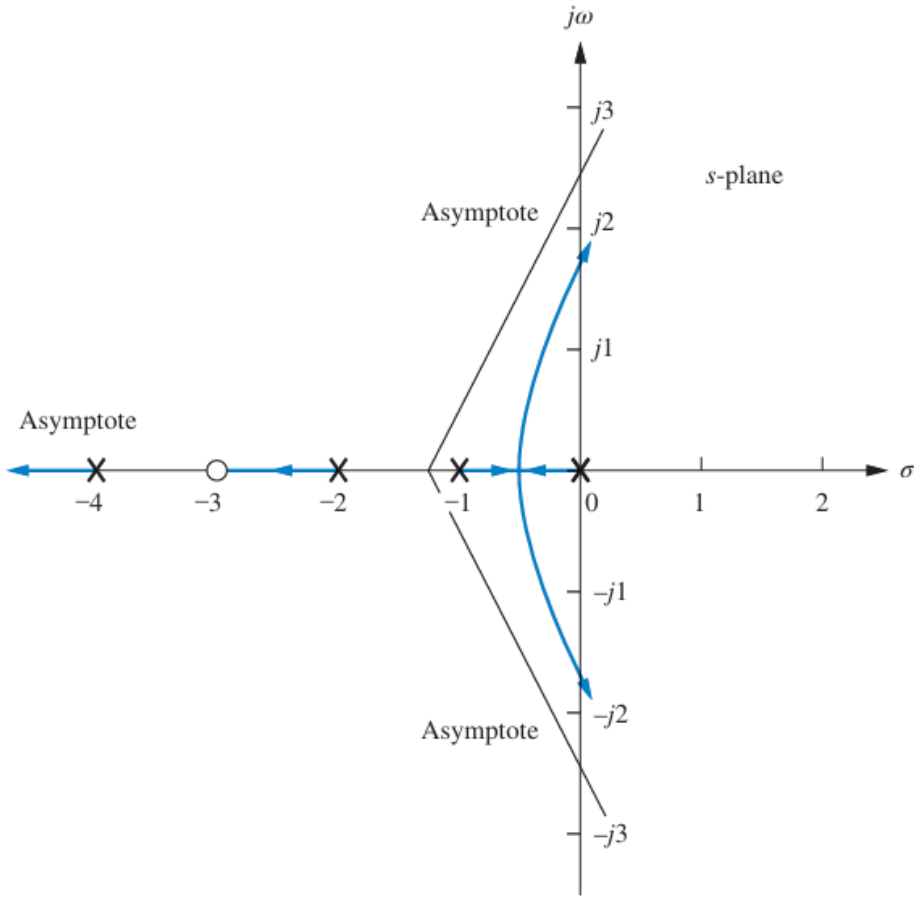


$$\Rightarrow \frac{G}{1+GH} = \frac{\frac{k(s+3)}{s(s+1)(s+2)(s+4)}}{1+\frac{k(s+3)}{s(s+1)(s+2)(s+4)}}$$

$$\frac{C(s)}{R(s)} = \frac{k(s+3)}{s(s+1)(s+2)(s+4)+k(s+3)}$$

$$s(s+1)(s+2)(s+4)+k(s+3)=0$$

$$1+\frac{k(s+3)}{s(s+1)(s+2)(s+4)}=0 \qquad \begin{pmatrix} m=1 \\ n=4 \end{pmatrix}$$



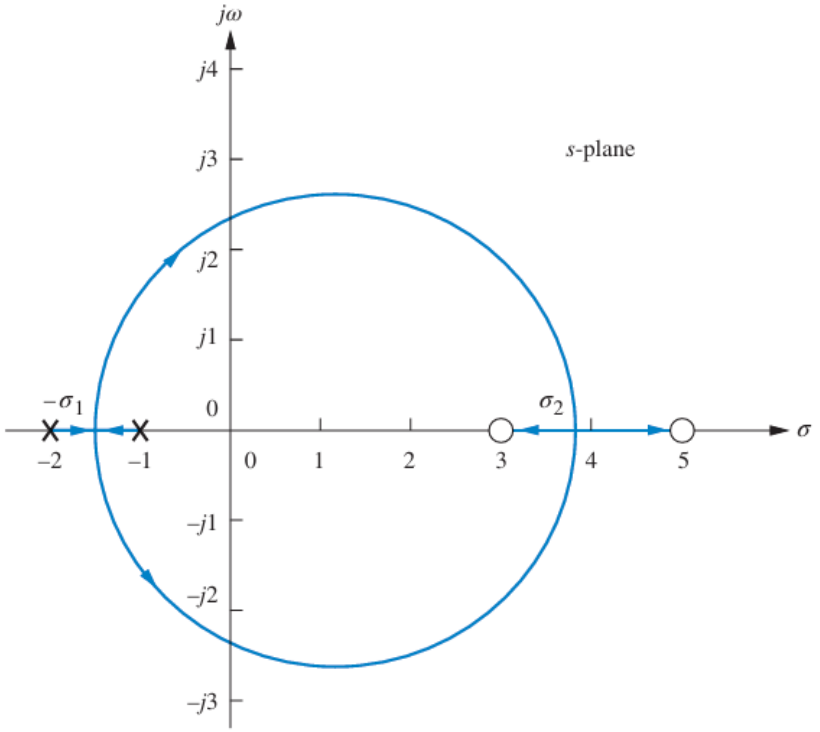
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근궤적 Loot Locus

$z = 3, 5$

$p = -1, -2$

$$KG(s)H(s) = \frac{K(s-3)(s-5)}{(s+1)(s+2)}$$ G_1H_1



$$|G_1(s)H_1(s)| = \frac{1}{|K|}$$

$$KG(s)H(s) = \frac{(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}$$

$$\frac{d}{ds}(G_1(s)H_1(s)) = 0$$

$$\frac{dK}{ds} = \frac{dG_1(s)H_1(s)/ds}{[G_1(s)H_1(s)]^2}$$

$$\sum_1^m \frac{1}{\sigma + z_i} = \sum_1^n \frac{1}{\sigma + p_i}$$

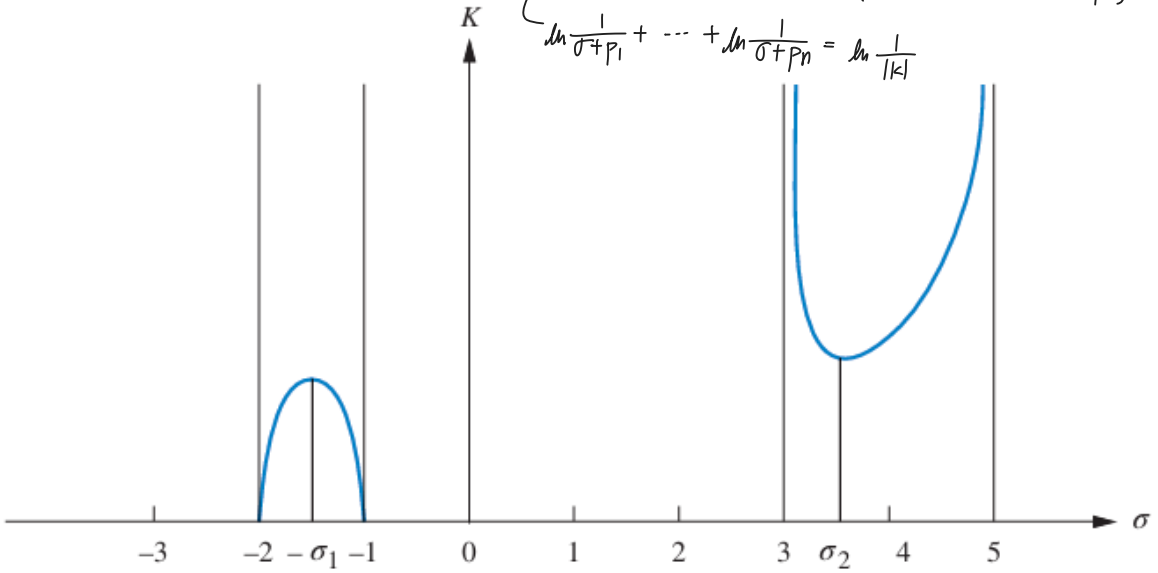
$$\frac{dK}{ds} = 0$$

이탈점 위치

$$\Rightarrow \ln \frac{|(\sigma+z_1)||(\sigma+z_2)|\cdots|(\sigma+z_m)|}{|(\sigma+p_1)|\cdots|(\sigma+p_n)|} = \ln \frac{1}{|K|}$$

$$\ln|\sigma+z_1| + \cdots + \ln|\sigma+z_m| - (\ln|\sigma+p_1| + \cdots + \ln|\sigma+p_n|) = -\ln|K|$$

$$\ln\frac{1}{\sigma+p_1} + \cdots + \ln\frac{1}{\sigma+p_n} = \ln\frac{1}{|K|}$$



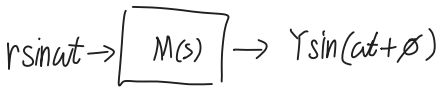
주파수 응답 소개

✓선형시불변 시스템

- 시스템의 정상상태 출력은 동일 주파수를 가지며 변화된 진폭과 위상을 갖음

$$r(t) = R\sin\omega t$$

$$y(t) = Y\sin(\omega t + \phi)$$



- 출력대비 입력에 대한 라플라스 변환

$$Y(s) = M(s)R(s)$$

- 정상상태 분석을 위해 s에 $j\omega$를 대입

$$Y(j\omega) = M(j\omega)R(j\omega)$$

주파수 응답 소개

✓선형시불변 시스템

- 입력과 출력 사이의 크기와 위상 관계

$$Y(j\omega) = M(j\omega)R(j\omega)$$

$$Y(j\omega) = |Y(j\omega)| \angle Y(j\omega)$$

$$|Y(j\omega)| = |M(j\omega)||R(j\omega)| \qquad \angle Y(j\omega) = \angle M(j\omega) + \angle R(j\omega)$$

$$r(t) = R\sin\omega t$$

$$y(t) = Y\sin(\omega t + \phi)$$



$$Y = R|M(j\omega_0)|$$

$$\phi = \angle M(j\omega_0)$$

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주파수 응답 소개

✓선형시불변 시스템

$A\sin\omega t \rightarrow \boxed{G(s)} \rightarrow$

$$G(s) = \frac{1}{(s+2)(s+8)} = \frac{1}{s^2+10s+16} = \frac{1}{s^2+2\zeta\omega_n s+\omega_n^2}$$

출력 : $Z(s) = \frac{A\omega}{s^2+\omega^2}\frac{1}{(s+2)(s+8)}$

$$\mathcal{L}[\sin\omega t] = \frac{\omega}{s^2+\omega^2}$$

$\frac{0}{s^2+\omega^2} + \frac{0}{s+2} + \frac{0}{s+8}$ — $0e^{-2t}$ $0e^{-8t}$ 시간에따라 없어질것

결국 애만남음

$$z(t) = \{\text{Transients} \rightarrow 0\} + A\frac{1}{\sqrt{(\omega^2+4)(\omega^2+64)}}\sin\left(\omega t + \tan^{-1}\frac{-10\omega}{(16-\omega^2)}\right)$$

$\omega = 2\pi f$

- $\omega = 0$: $\frac{A}{\sqrt{4\times64}}$
- $\omega = 1$: $\frac{A}{\sqrt{5\times65}}$

$$z(t) = Z\sin(\omega t+\phi)$$
$$= A|G(j\omega)|\angle G(j\omega)$$
$$= AM\angle\phi$$
$$= Z\angle\phi$$

$$M = |G(j\omega)| = \frac{1}{\sqrt{(\omega^2+4)(\omega^2+64)}}$$

$$\phi = \angle G(j\omega) = \tan^{-1}\frac{-10\omega}{(16-\omega^2)}$$

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주파수 응답 소개

✓선형시불변 시스템

$$M = |G(j\omega)| = \frac{1}{\sqrt{(\omega^2+4)(\omega^2+64)}}$$

$$G(j\omega) = \frac{1}{(\omega j+2)(\omega j+8)}$$

$$\phi = \angle G(j\omega) = \tan^{-1}\frac{-10\omega}{(16-\omega^2)}$$

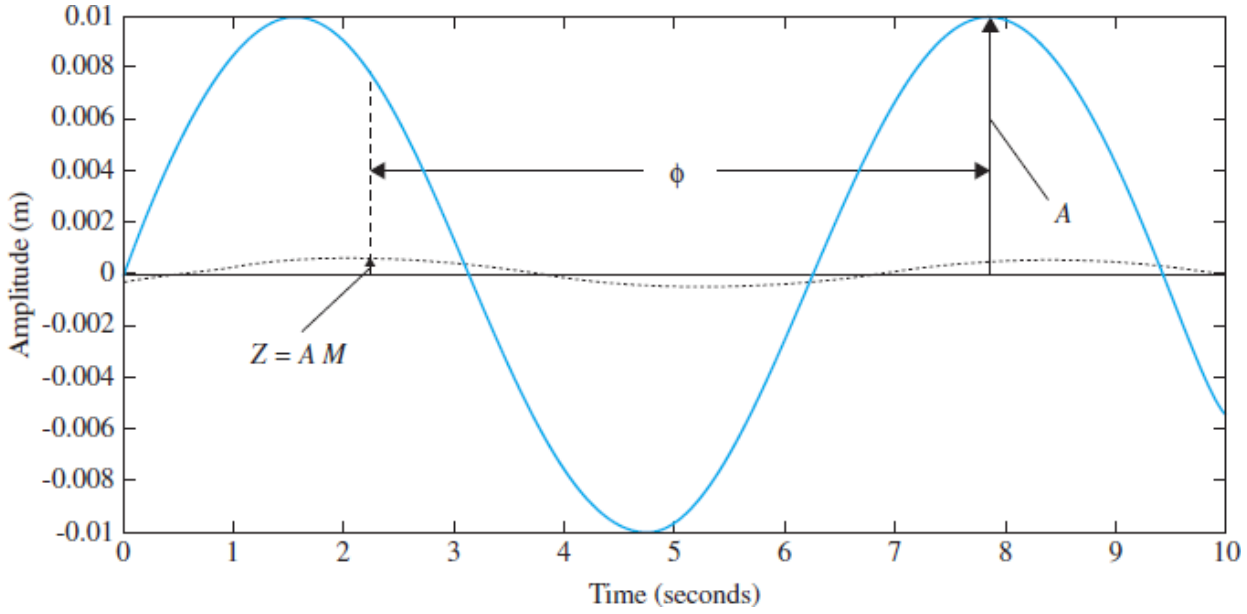


Figure 10-3 Graphical representation of input $A\ sin(\omega t)$ and output $z(t) = Z\ \sin(\omega t + \phi)$ for $A = 0.01$ and $\omega = 1$ rad/s.

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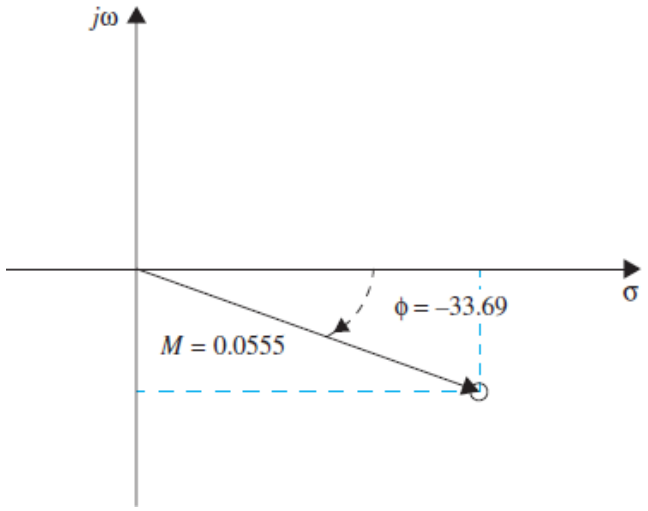
주파수 응답 소개

✓선형시불변 시스템

$$M = |G(j\omega)| = \frac{1}{\sqrt{(\omega^2+4)(\omega^2+64)}}$$

$$G(j\omega) = \frac{1}{(\omega j+2)(\omega j+8)}$$

$$\phi = \angle G(j\omega) = \tan^{-1}\frac{-10\omega}{(16-\omega^2)}$$



ω = 1 rad/s

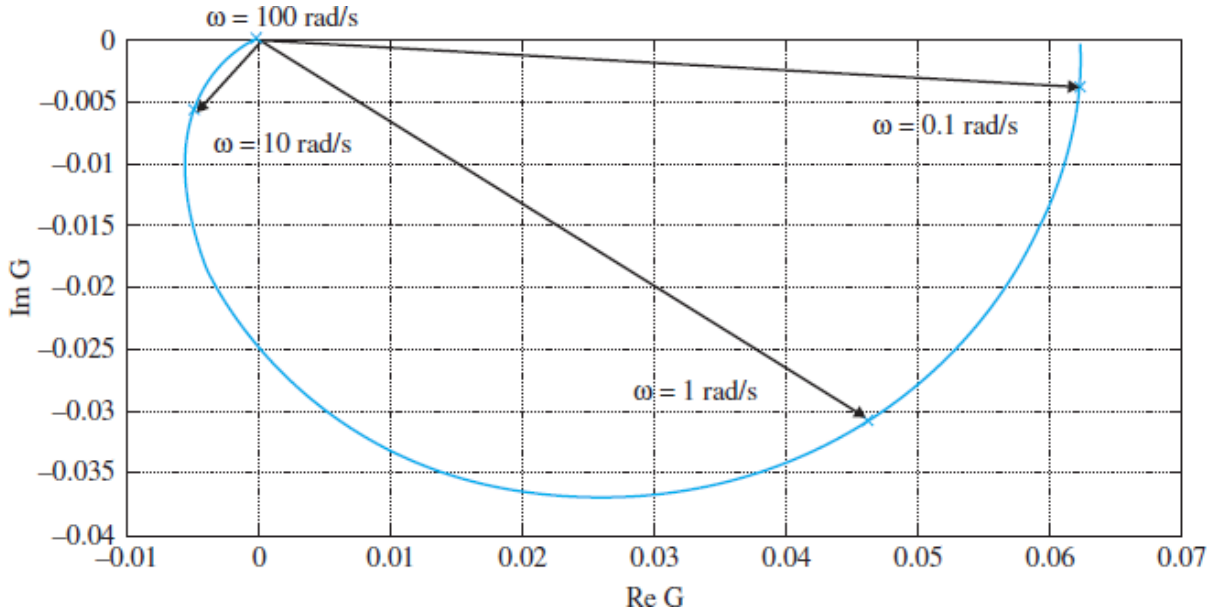


Figure 10-4 Polar representation of the frequency response function $G(j\omega) = \frac{1}{(\omega j+2)(\omega j+8)}$ for w varying from 0.1 to 100 rad/s.

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주파수 응답 소개

✓선형시불변 시스템

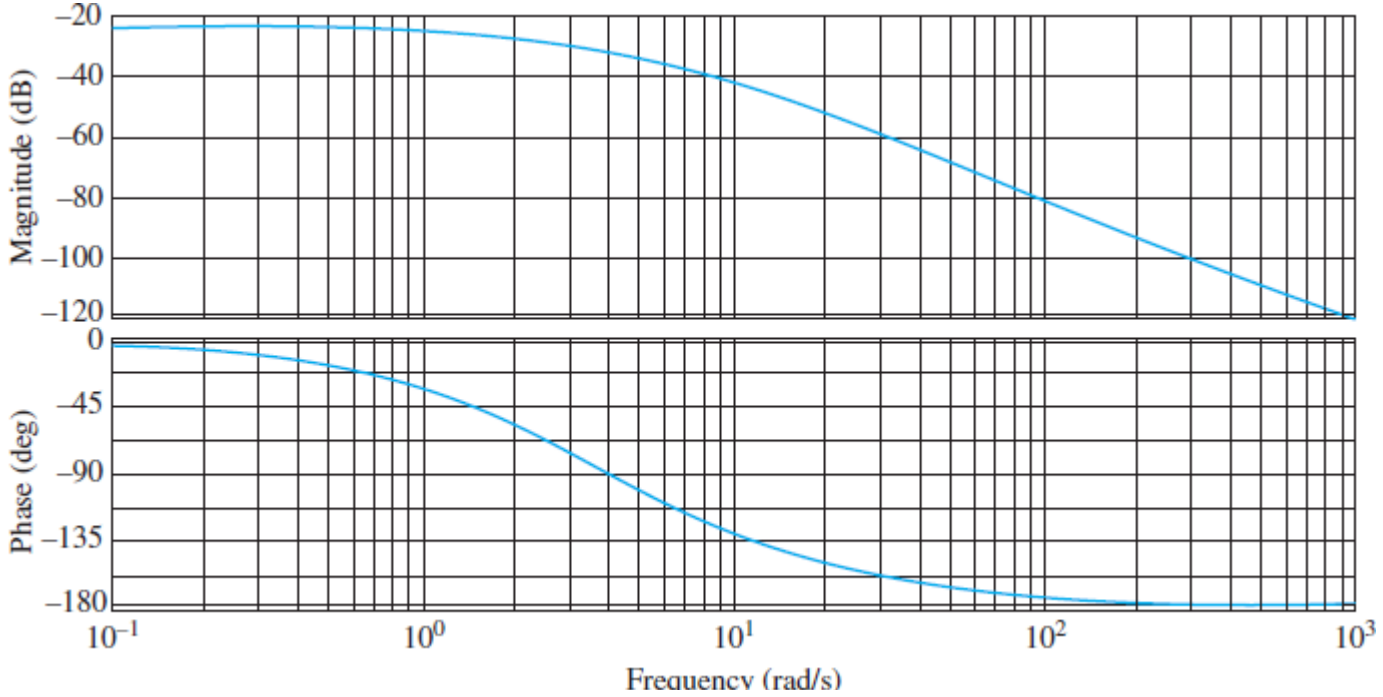


Figure 10-5 Magnitude and phase frequency response plots for $G(j\omega)=\dfrac{1}{(\omega j+2)(\omega j+8)}$ for ω varying from 0.1 to 100 rad/s.

$$M=|G(j\omega)|=\frac{1}{\sqrt{(\omega^2+4)(\omega^2+64)}}$$

$$\phi=\angle G(j\omega)=\tan^{-1}\frac{-10\omega}{(16-\omega^2)}$$

ω rad/s	M	ϕ Degrees
0.1	0.0625	–3.58
1	0.0555	–33.69
10	0.0077	–130.03
100	0.0001	–174.28

주파수 응답 소개

✓제어 시스템의 주파수 영역 해석

$$G(s)=\frac{10}{s(s+1)(s+2)}$$

$$\lim_{\omega\to 0}|G(j\omega)|=\lim_{\omega\to 0}\frac{5}{\omega}=\infty$$

$$\lim_{\omega\to 0}\angle G(j\omega)=\lim_{\omega\to 0}\angle 5/j\omega=-90^\circ$$

$$\lim_{\omega\to \infty}|G(j\omega)|=\lim_{\omega\to \infty}\frac{10}{\omega^3}=0$$

$$G(j\omega)=\frac{10(-j\omega)(-j\omega+1)(-j\omega+2)}{j\omega(j\omega+1)(j\omega+2)(-j\omega)(-j\omega+1)(-j\omega+2)}$$

$$G(j\omega)=\mathrm{Re}[G(j\omega)]+j\,\mathrm{Im}[G(j\omega)]=\frac{-30}{9\omega^2+(2-\omega^2)^2}-\frac{j10(2-\omega^2)}{9\omega^3+\omega(2-\omega^2)^2}$$

$$\mathrm{Im}[G(j\omega)]=0$$

$$\mathrm{Re}[G(j\omega)]=0$$

$$G(\pm j\sqrt{2})=-5/3$$

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주파수 응답 소개

✓제어 시스템의 주파수 영역 해석

$$G(s)=\frac{10}{s(s+1)(s+2)}$$

$$G(j\omega)=\mathrm{Re}[G(j\omega)]+j\mathrm{Im}[G(j\omega)]=\frac{-30}{9\omega^2+(2-\omega^2)^2}-\frac{j10(2-\omega^2)}{9\omega^3+\omega(2-\omega^2)^2}$$

$$G(\pm j\sqrt{2})=-5/3$$

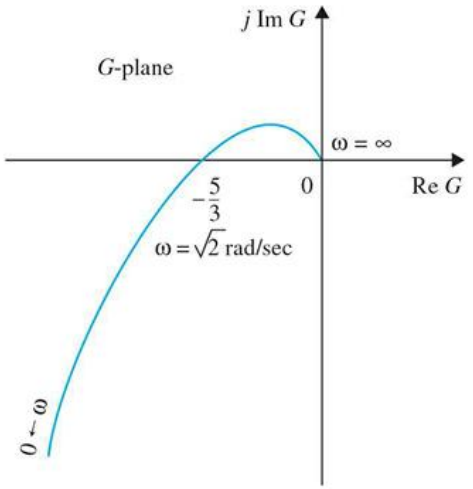
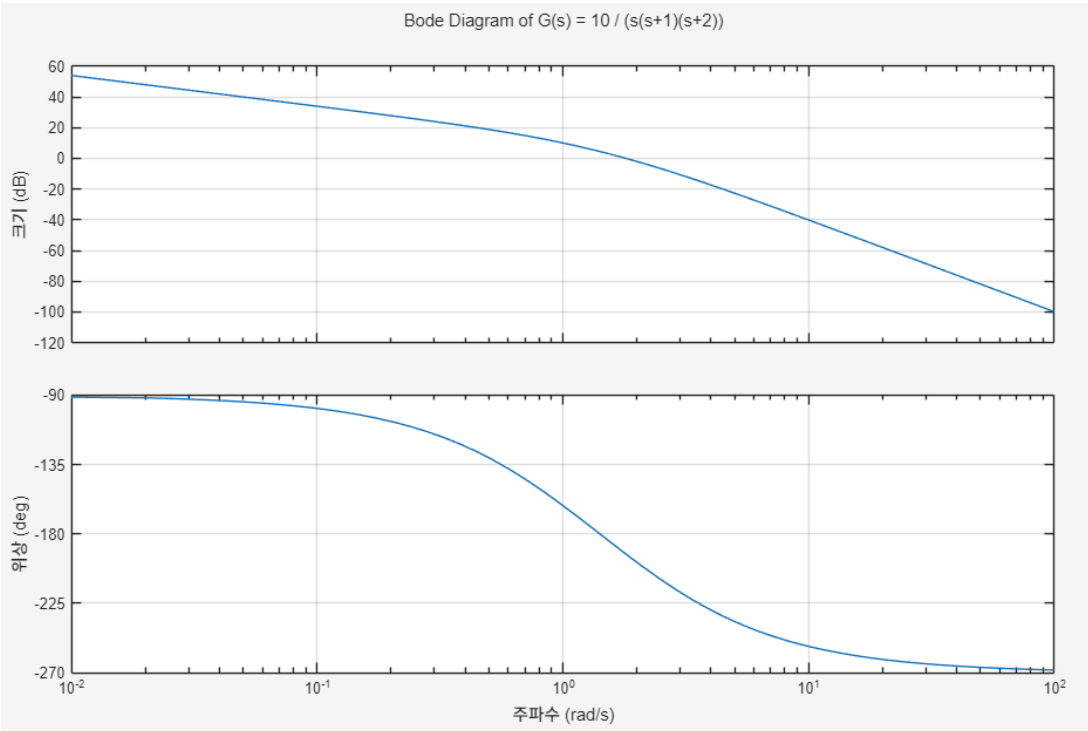


Figure 10-6 Polar plot of $G(s)=\frac{10}{s(s+1)(s+2)}$.



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주파수 응답 소개

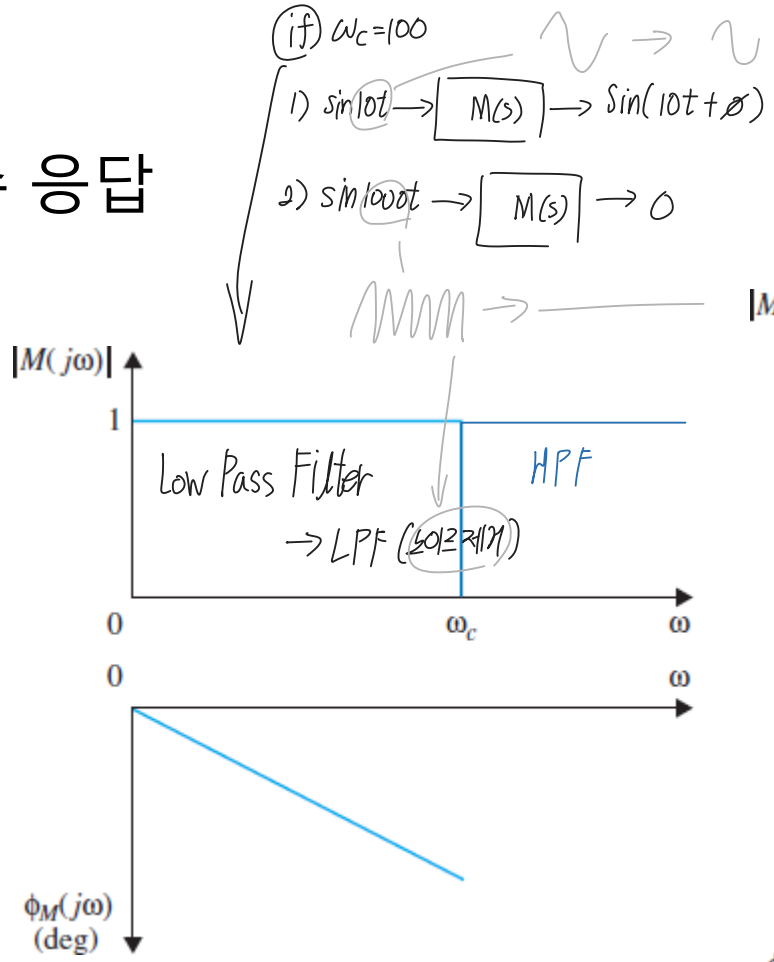
✓폐루프시스템의 주파수 응답

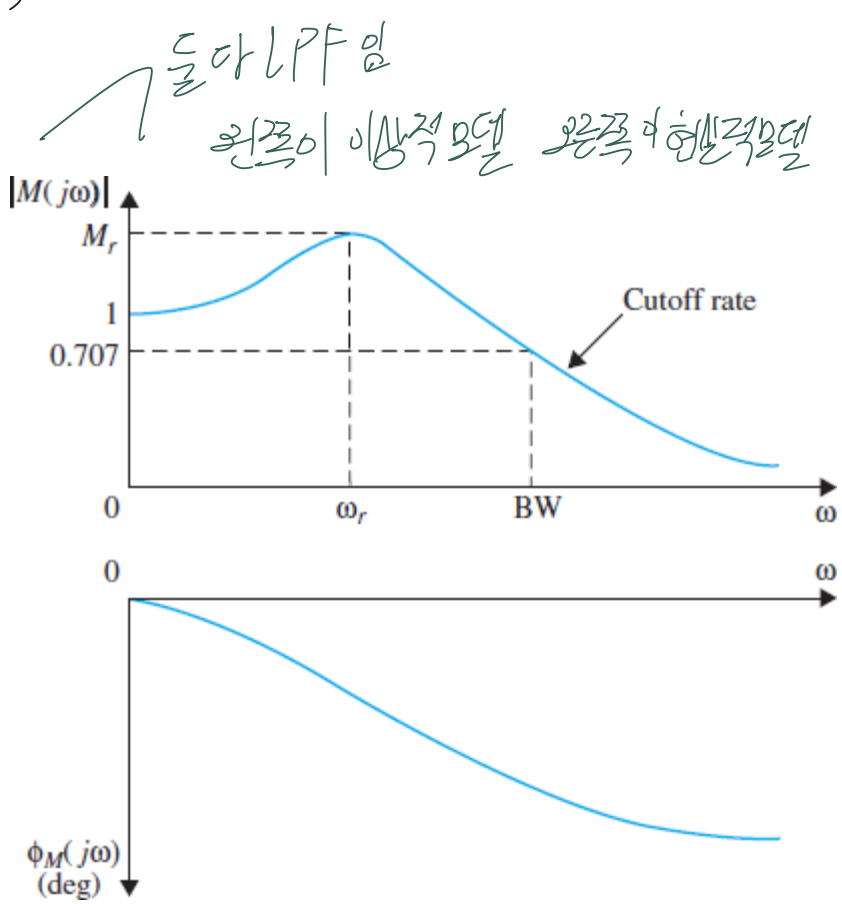
$$M(s)=\frac{Y(s)}{R(s)}=\frac{G(s)}{1+G(s)H(s)}$$

$$M(j\omega)=\frac{Y(j\omega)}{R(j\omega)}=\frac{G(j\omega)}{1+G(j\omega)H(j\omega)}$$

$$M(j\omega)=\left|M(j\omega)\right|\angle M(j\omega)$$

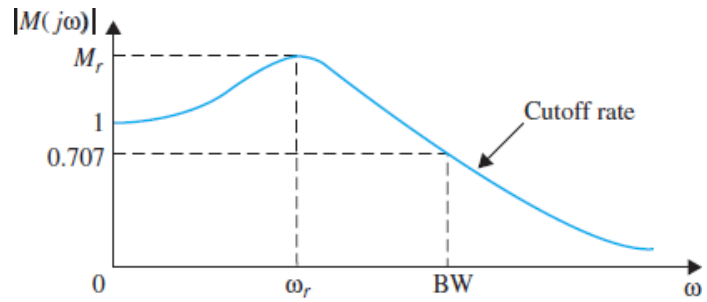
$$M(j\omega)=\mathrm{Re}[M(j\omega)]+j\mathrm{Im}[M(j\omega)]$$





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주파수 응답 소개



✓주파수영역 사양

- 첨두공진치 M_r
  - $|M(j\omega)|$의 최대값
  - M_r의 크기는 안정한 피드백 제어시스템의 상대적 안정도의 한 척도
  - 큰 M_r은 시간영역에서 계단응답의 큰 최대오버슈트에 대응
- 공진주파수 ω_r
  - 첨두공진치 M_r이 발생하는 주파수
- 대역폭 BW
  - $|M(j\omega)|$의 크기가 주파수가 0일 때 값의 70.7% 또는 3dB 떨어질 때의 주파수
- 차단율
  - 높은 주파수에서 $|M(j\omega)|$의 기울기

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표준형 2차 시스템의 M_r, ω_r 및 대역폭

✓첨두공진치와 공진주파수 Resonant peak and resonant frequency

- 표준형 2차 시스템의 폐루프 전달함수

$$M(s)=\frac{Y(s)}{R(s)}=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$s=j\omega$

$$M(j\omega)=\frac{Y(j\omega)}{R(j\omega)}=\frac{\omega_n^2}{(j\omega)^2+2\zeta\omega_n(j\omega)+\omega_n^2}$$

$$=\frac{1}{1+j2(\omega/\omega_n)\zeta-(\omega/\omega_n)^2}$$

$$M(ju)=\frac{1}{1+j2u\zeta-u^2} \qquad u=\omega/\omega_n$$

$$|M(ju)|=\frac{1}{[(1-u^2)^2+(2\zeta u)^2]^{1/2}}$$

$$\angle M(ju)=\phi_M(ju)=-\tan^{-1}\frac{2\zeta u}{1-u^2}$$

$$\frac{d|M(ju)|}{du}=-\frac{1}{2}[(1-u^2)^2+(2\zeta u)^2]^{-3/2}(4u^3-4u+8u\zeta^2)=0$$

$$4u^3-4u+8u\zeta^2=4u(u^2-1+2\zeta^2)=0$$

$$u_r=\sqrt{1-2\zeta^2}$$

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표준형 2차 시스템의 M_r, ω_r 및 대역폭

✓첨두공진치와 공진주파수 Resonant peak and resonant frequency

- 표준형 2차 시스템의 폐루프 전달함수

$$M(s)=\frac{Y(s)}{R(s)}=\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} \qquad |M(ju)|=\frac{1}{[(1-u^2)^2+(2\zeta u)^2]^{1/2}}$$

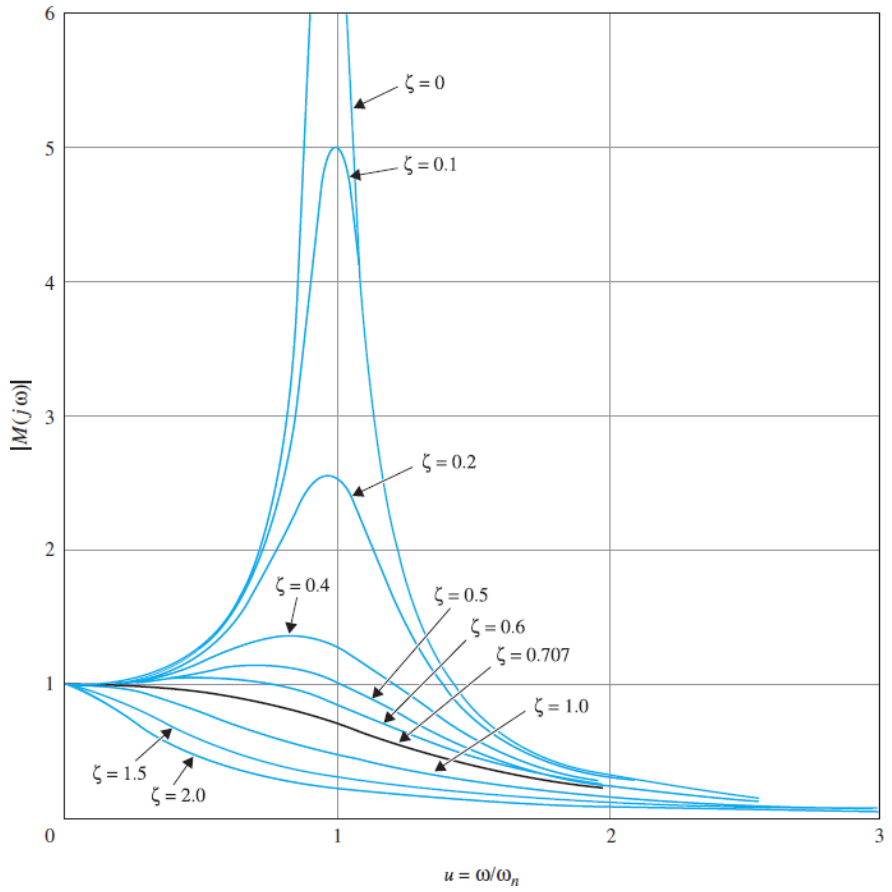
$$u_r=\sqrt{1-2\zeta^2}$$

$\zeta \downarrow \rightarrow M_r \uparrow$

$$\omega_r=\omega_n\sqrt{1-2\zeta^2}$$

$\% OS \uparrow$

$$M_r=\frac{1}{2\zeta\sqrt{1-\zeta^2}} \quad \zeta \le 0.707$$



표준형 2차 시스템의 M_r, ω_r 및 대역폭

✓첨두공진치와 공진주파수 Resonant peak and resonant frequency

- 표준형 2차 시스템의 폐루프 전달함수

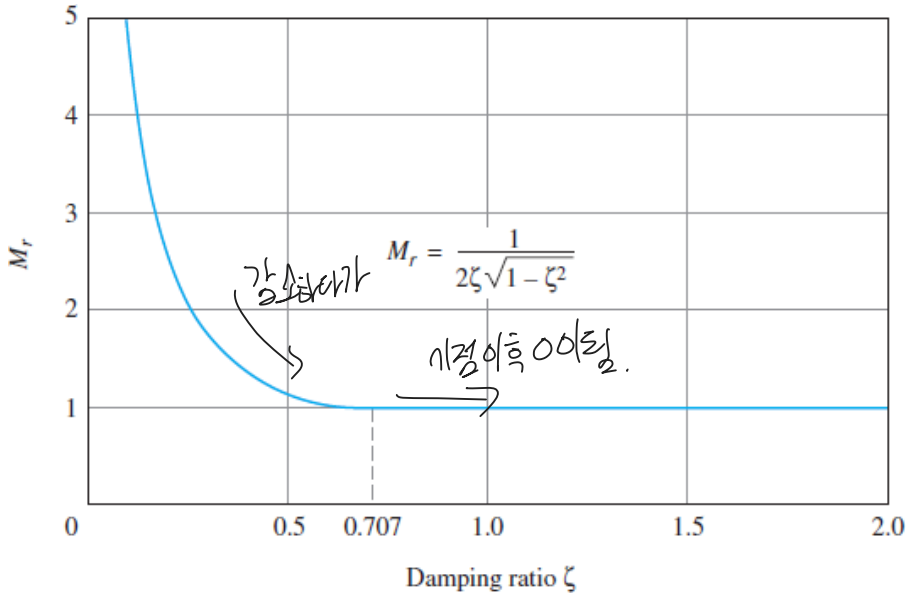


Figure 10-11 M_r versus damping ratio for the prototype second-order system.

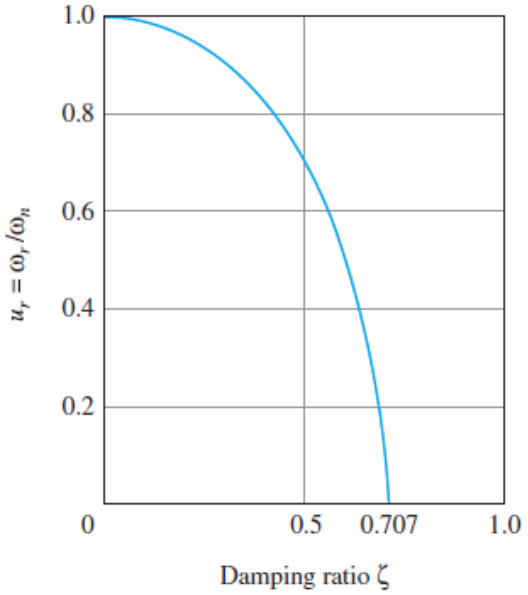


Figure 10-12 Normalized resonant frequency versus damping ratio for the prototype second-order system. $u_r = \sqrt{1-2\zeta^2}$.

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표준형 2차 시스템의 M_r, ω_r 및 대역폭

✓첨두공진치와 공진주파수 Resonant peak and resonant frequency

- 표준형 2차 시스템의 폐루프 전달함수

$$|M(ju)| = \frac{1}{[(1-u^2)^2 + (2\zeta u)^2]^{1/2}} = \frac{1}{\sqrt{2}}$$

$$[(1-u^2)^2 + (2\zeta u)^2]^{1/2} = \sqrt{2}$$

$$u^2 = (1-2\zeta^2) \pm \sqrt{4\zeta^4 - 4\zeta^2 + 2}$$

$$\text{BW} = \omega_n \left[(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2} \right]^{1/2}$$

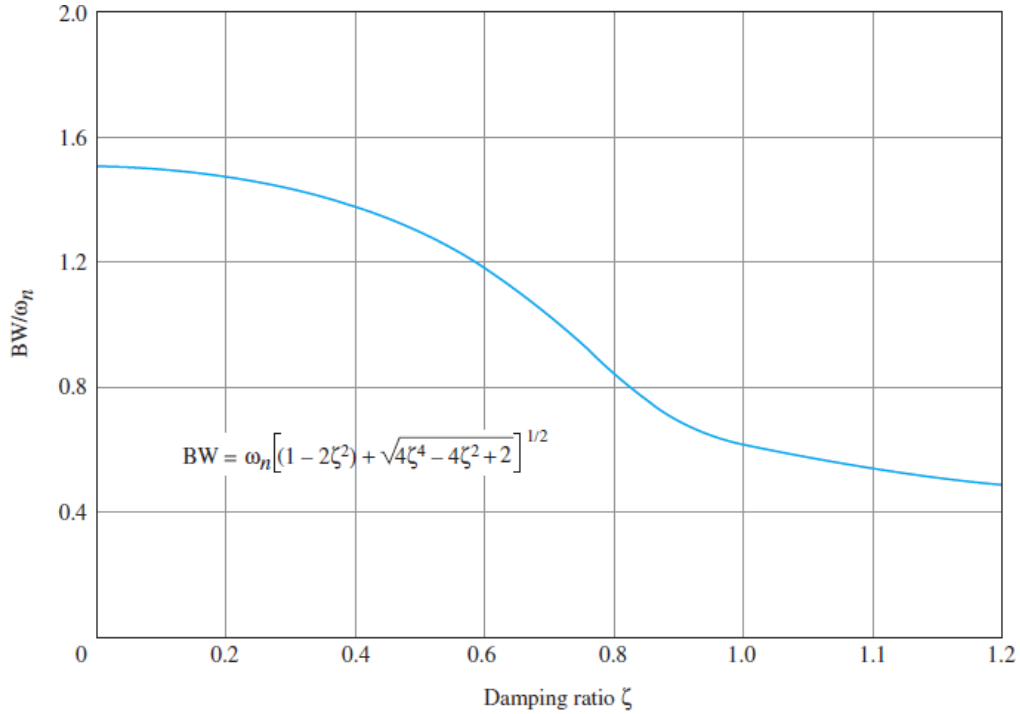


Figure 10-13 Bandwidth/ω_n versus damping ratio for the prototype second-order system.

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표준형 2차 시스템의 M_r, ω_r 및 대역폭

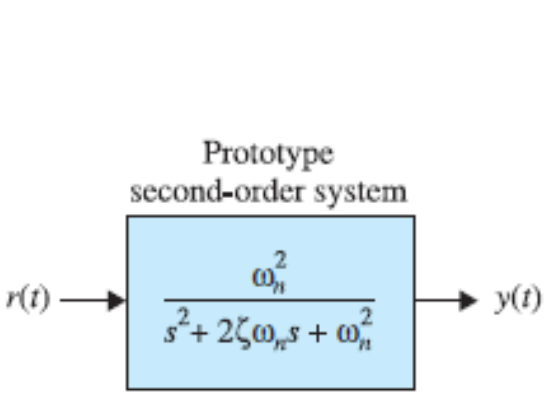
✓2차 시스템의 시간영역 응답과 주파수영역 특성 사이 관계

- 첨두공진치 M_r은 ζ에 의존, ζ가 0일때 M_r은 무한대, ζ가 음이면 시스템은 불안정하고 M_r의 값은 무의미, ζ가 증가하면 M_r은 감소
- $\zeta \geq 0.707$에서 $M_r = 1, \omega_r = 0$, 최대오버슈트 역시 ζ에 의존
- 대역폭 BW는 ω_n에 비례, 고정된 ω_n에서 BW는 ζ가 증가함에 따라 감소, 단위계단응답에 대해서는 ω_n이 감소할 때 상승시간이 증가. BW오 상승시간은 서로 반비례 관계
- $0 \leq \zeta \leq 0.707$에서 BW와 M_r은 서로 반비례

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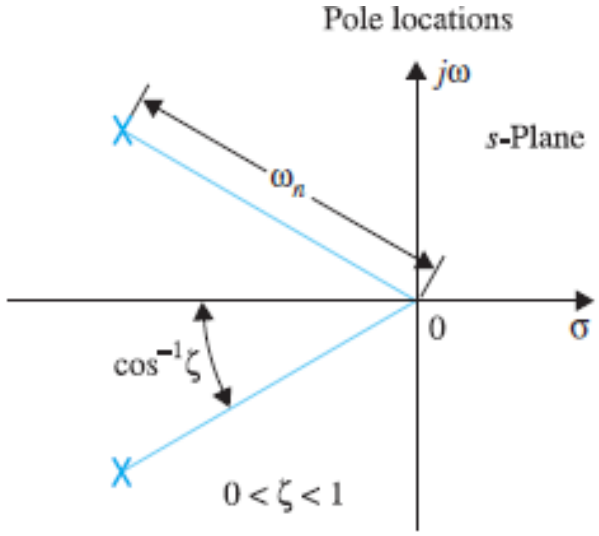
표준형 2차 시스템의 M_r, ω_r 및 대역폭

✓2차 시스템의 시간영역 응답과 주파수영역 특성 사이 관계



As ω_n gets larger, pole distance from origin gets larger.

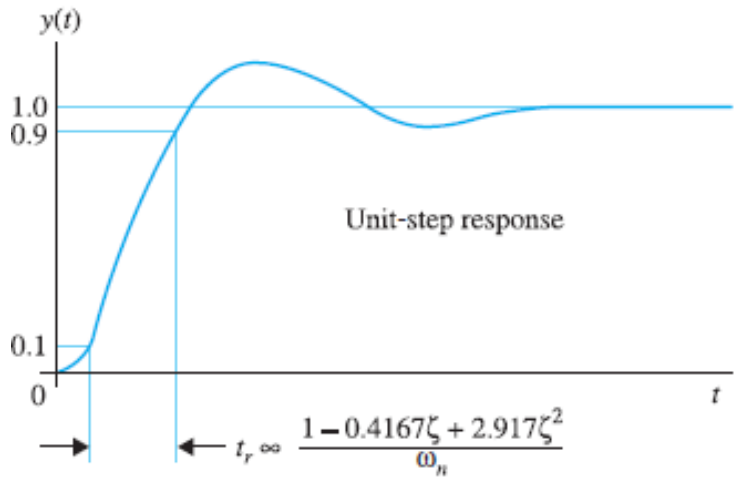
As ζ gets larger, angular distance from negative real axis gets smaller.



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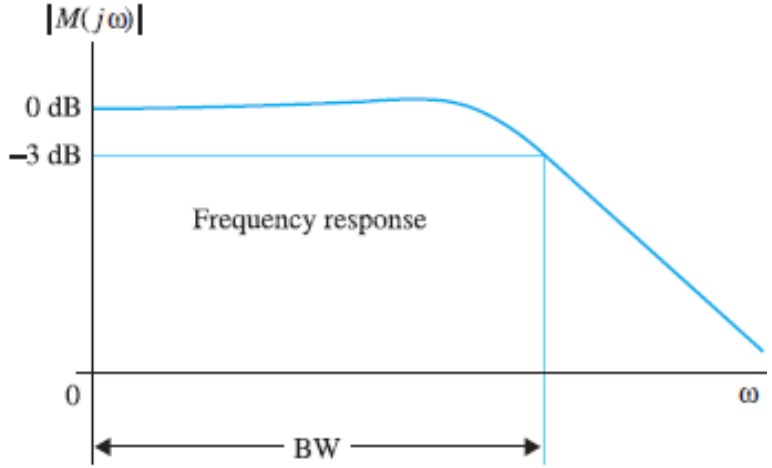
표준형 2차 시스템의 M_r, ω_r 및 대역폭

✓2차 시스템의 시간영역 응답과 주파수영역 특성 사이 관계



As ω_n gets larger, t_r gets smaller and the system responds faster.

As ζ gets larger, t_r gets larger and the system responds slower.



$$BW = \omega_n\left[(1-2\zeta^2) + \sqrt{4\zeta^4 - 4\zeta^2 + 2}\right]^{1/2}$$

As ω_n gets larger, BW gets larger.

As ζ gets larger, BW gets smaller.

Bandwidth and rise time are inversely proportional.

Therefore, the larger the bandwidth is, the faster the system will respond.

Increasing ω_n increases BW and decreases t_r.

Increasing ζ decreases BW and increases t_r.

전방경로전달함수에 극과 영점을 추가하는 효과

✓전방경로 전달함수에 영점 추가의 효과

- 폐루프 시스템의 대역폭을 증가시키는 일반적인 효과

$$G(s)=\frac{\omega_n^2}{s(s+2\zeta\omega_n)}$$

Add a zero at $s = -1/T$

$$G(s)=\frac{\omega_n^2(1+Ts)}{s(s+2\zeta\omega_n)}$$

$$M(s)=\frac{\omega_n^2(1+Ts)}{s^2+(2\zeta\omega_n+T\omega_n^2)s+\omega_n^2}$$

$$\mathrm{BW}=\left(-b+\frac{1}{2}\sqrt{b^2+4\omega_n^4}\right)^{1/2}$$

$$b=4\zeta^2\omega_n^2+4\zeta\omega_n^3T-2\omega_n^2-\omega_n^4T^2$$

$\zeta = 0.707$ $\omega_n = 1$

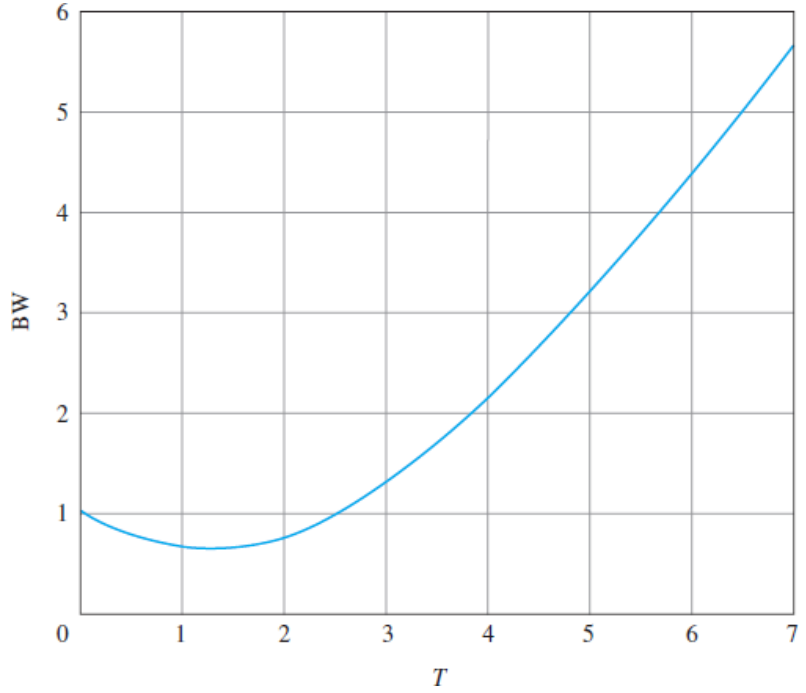


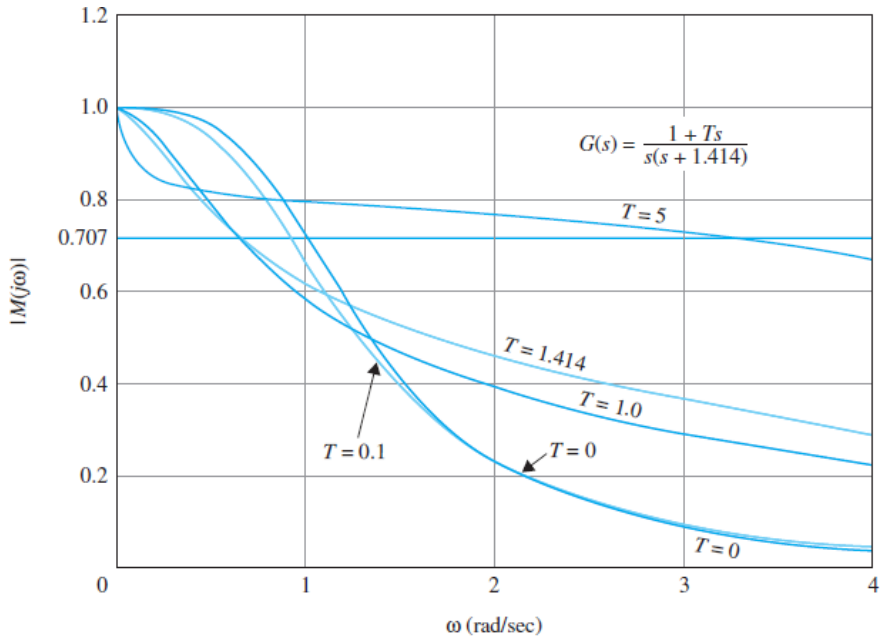
Figure 10-15 Bandwidth of a second-order system with open-loop transfer function $G(s) = (1 + Ts)/[s(s + 1.414)]$.

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전방경로전달함수에 극과 영점을 추가하는 효과

✓전방경로 전달함수에 영점 추가의 효과

- 폐루프 시스템의 대역폭을 증가시키는 일반적인 효과



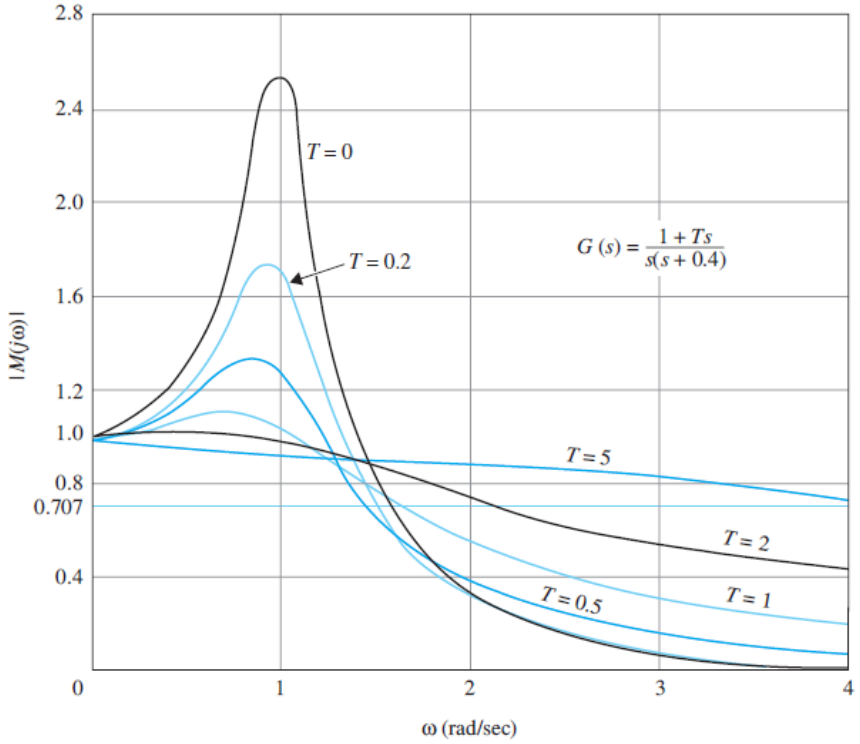


Figure 10-16 Magnification curves for the second-order system with the forward-path transfer function $G(s)$ in Eq. (10-46). (*a*) $\omega n = 1$, $\zeta = 0.707$ (*b*) $\omega n = 1$, $\zeta = 0.2$.

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전방경로전달함수에 극과 영점을 추가하는 효과

✓전방경로 전달함수에 영점 추가의 효과

- 폐루프 시스템의 대역폭을 증가시키는 일반적인 효과

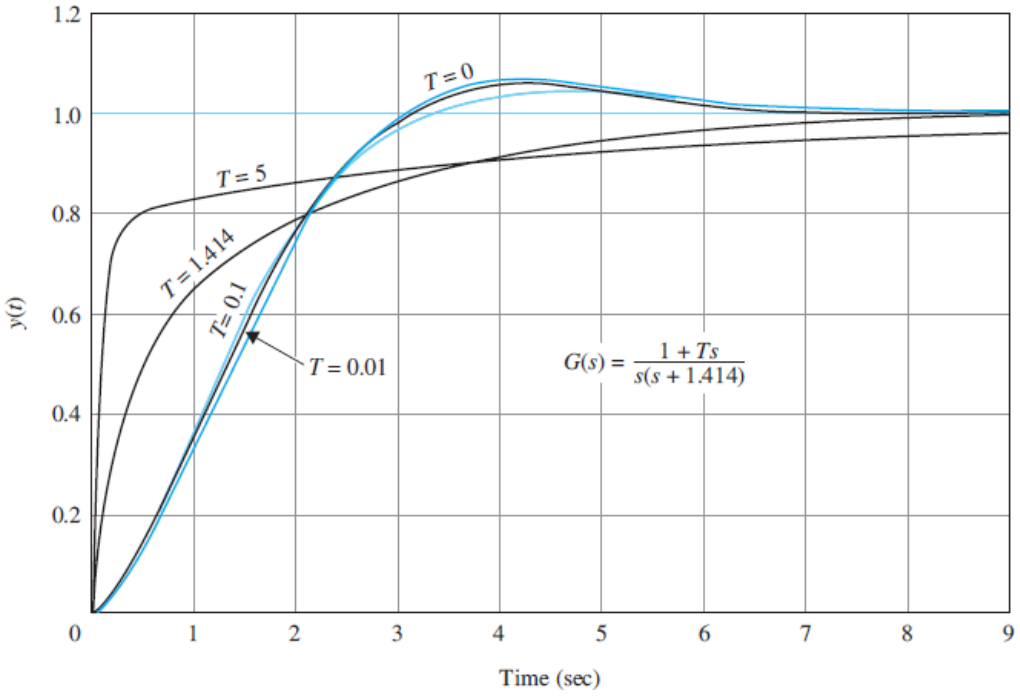


Figure 10-17 Unit-step responses of a second-order system with a forward-path transfer function $G(s)$.

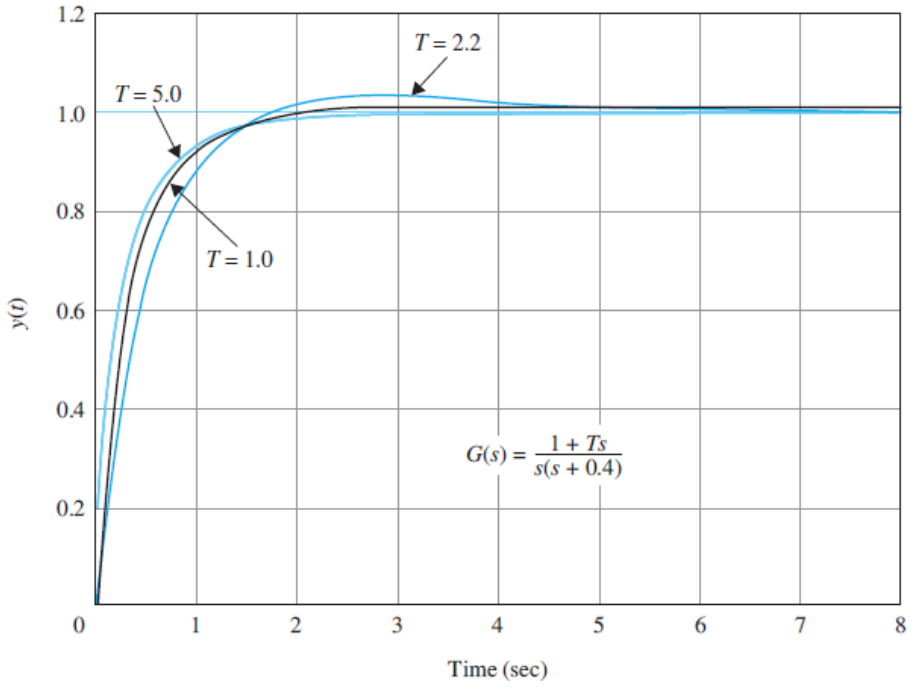


Figure 10-18 Unit-step responses of a second-order system with a forward-path transfer function $G(s)$.

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전방경로전달함수에 극과 영점을 추가하는 효과

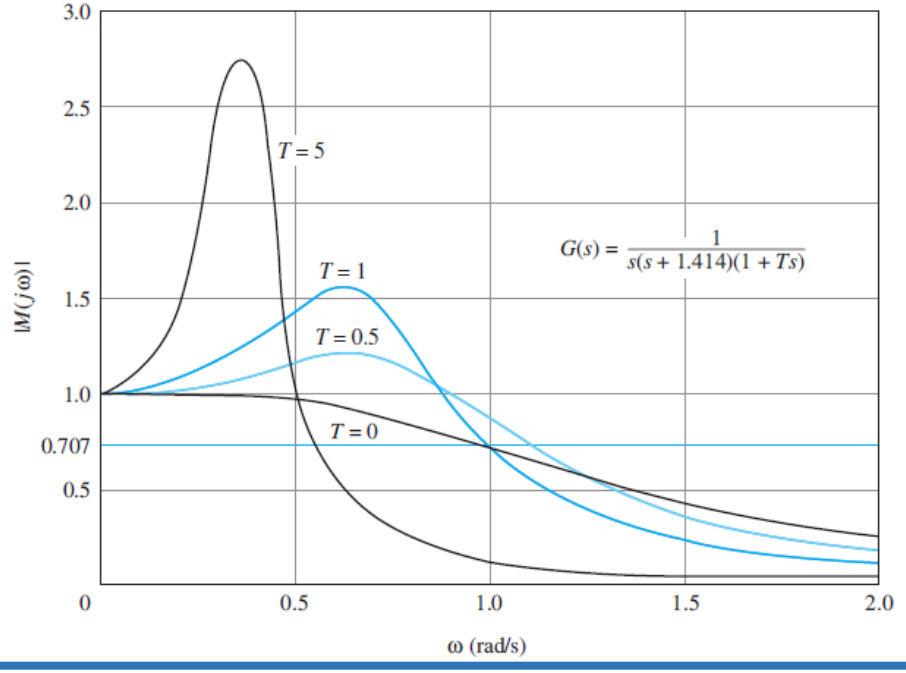
✓전방경로 전달함수에 극 추가의 효과

- 폐루프시스템은 덜 안정되고 대역폭은 감소
- 상승시간은 대역폭이 감소할 때 증가
- M_r의 큰 값은 단위계단응답의 더 큰 최대오버슈트에 대응

영점: 대역폭↑
극점: 대역폭↓

$$G(s)=\frac{\omega_n^2}{s(s+2\zeta\omega_n)}$$ Add a pole at $s = -1/T$

$$G(s)=\frac{\omega_n^2}{s(s+2\zeta\omega_n)(1+Ts)}$$



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전방경로전달함수에 극과 영점을 추가하는 효과

✓전방경로 전달함수에 극 추가의 효과

- 상승시간은 대역폭이 감소할 때 증가
- M_r의 큰 값은 단위계단응답의 더 큰 최대오버슈트에 대응

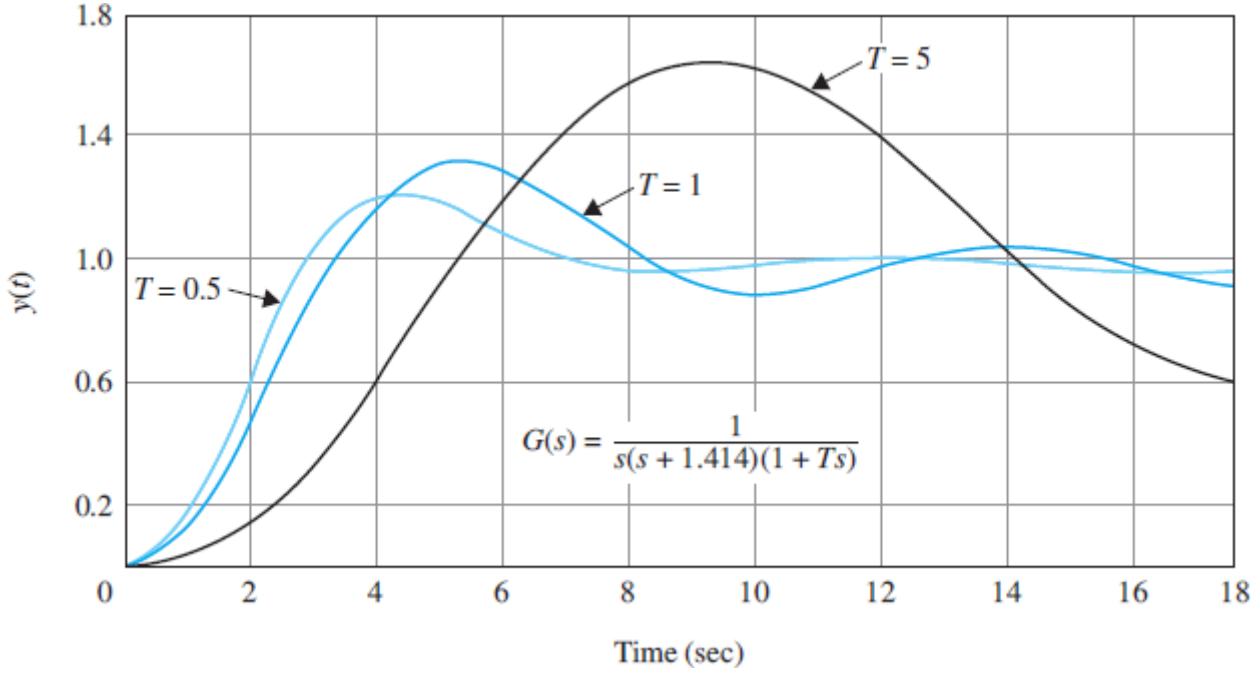


Figure 10-20 Unit-step responses of a third-order system with a forward-path transfer function $G(s)$.